

# DAY 09: DOCKER VOLUMES, DOCKER NETWORKING & MAVEN BUILD TOOL

## Cloud Computing & DevOps Master Curriculum

### 1. Docker Volumes & Data Persistence

By default, data created inside a container is ephemeral and deleted when the container is removed.

**Docker Volumes** provide persistent storage managed entirely by Docker on the host filesystem.

```
# Create a Docker volume
docker volume create app-data-vol

# Run container with volume mounted
docker run -d -v app-data-vol:/app/data --name app-service my-app:latest

# Inspect volume path
docker volume inspect app-data-vol
```

### 2. Docker Networking Drivers

| Driver           | Description                                                                                  |
|------------------|----------------------------------------------------------------------------------------------|
| Bridge (Default) | Isolated private network on single host. Containers communicate via container IP or alias.   |
| Host             | Removes network isolation between container and Docker host (shares host IP/ports directly). |
| Overlay          | Enables multi-host networking across multiple Docker Swarm / node daemons.                   |
| None             | Disables all container networking (complete isolation).                                      |

### 3. Apache Maven & Containerizing Java Applications

**Apache Maven** is a build automation tool used primarily for Java projects. It manages dependencies using `pom.xml` and builds deployable JAR/WAR artifacts.

**Multi-stage Dockerfile for Maven Java App:**

```
# Stage 1: Build Application
FROM maven:3.8.6-openjdk-11 AS build
WORKDIR /app
COPY pom.xml .
COPY src ./src
RUN mvn clean package -DskipTests

# Stage 2: Run Application
FROM openjdk:11-jre-slim
WORKDIR /app
COPY --from=build /app/target/my-app-1.0.jar app.jar
EXPOSE 8080
```

```
ENTRYPOINT ["java", "-jar", "app.jar"]
```

#### KEY TAKEAWAYS & SUMMARY

- ✓ Docker volumes persist application state beyond container lifecycles.
- ✓ Bridge networking isolates container traffic; Overlay networking links containers across multiple servers.
- ✓ Multi-stage Docker builds reduce final production image size significantly by excluding build tools like Maven.